



Efficient shapley-based data valuation for federated trajectories

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■ 1 Introduction

The widespread adoption of GPS-enabled vehicles, bicycles, and mobile devices has made trajectory data a cornerstone of location-based services (LBS). When users share their movement traces, the aggregated data enables a wide range of applications. For example, vehicular trajectories support the construction of high-definition (HD) maps for autonomous navigation. Cycling trajectories help riders assess the safety and condition of bike lanes. Pedestrian traces guide hikers to select reliable routes [1]. In short, rich trajectory data improves situational awareness and decision-making for everyone in the ecosystem.

However, in practice, trajectories are collected and owned by different entities, and data sharing remains limited due to the lack of effective incentives. Data owners are typically unwilling to contribute their trajectories unless fairly compensated for the value their data provides. To address this challenge, it is crucial to develop a mechanism that fairly and efficiently quantifies the value of each contributor's trajectories, thereby enabling trustworthy incentives in federated setting [2].

To this end, we adopt the Shapley value, a classic cooperative game-theoretic concept known for its fairness properties, and formally define the problem of Shapley-based data valuation for federated trajectory datasets. We prove that the problem is #P-complete, motivating the need for efficient approximations. Accordingly, we propose an efficient Shapley value estimation framework tailored for trajectory datasets. Our approach is based on the following insights: (i) The utility function can be naturally decomposed across road segments and time slots, allowing more efficient Shapley value computation over smaller, independent units. (ii) Segment-time pairs that exhibit similar coverage patterns yield similar Shapley values. Hence it is unnecessary to compute the value for every pair. Accordingly, we propose a clustering-based framework that identifies and shares computations among segment-time pairs with similar coverage patterns, thus drastically reducing the computation overhead. Furthermore, we leverage permutation invariance and coverage saturation, two features in Shapley value of

trajectory dataset to improve the estimation accuracy. Extensive experiments on real-world datasets show that our method achieves up to $15.2\times$ speedup over existing baselines while maintaining high accuracy.

■ 2 Problem definition

2.1 Trajectory data and its coverage over road network

Definition 1 (Raw trajectory). Point p is defined by a timestamp $p.ts$ and the geo-location $p.loc$. A raw trajectory T_{raw} is defined as a time-ordered sequence of $|T_{raw}|$ points, i.e., $T_{raw} = \{p_1, p_2, \dots, p_{|T_{raw}|}\}$.

Due to GPS noise and other inaccuracies, raw trajectories often deviate from the actual road network. A common preprocessing step is map matching [3], which aligns raw trajectories to the road network. We define the road network and map-matched trajectories as follows:

Definition 2 (Road network). A road network is a graph $G = (V, E)$, where V is a set of vertices v representing road intersections and terminal points, and E is a set of edges e representing road segments.

Definition 3 (Map-matched trajectory). Given a raw trajectory T_{raw} and a road network G , a map-matched trajectory T is a sequence of road segments in G such that each GPS point in T_{raw} is projected onto the corresponding segment in G . Specifically, T is defined as $e_1 \rightarrow e_2 \rightarrow \dots \rightarrow e_{|T|}$, where each segment e_i is attached with a time interval indicating the period during which the trajectory traverses e_i .

In the following, we assume that all trajectories have already been map-matched to the road network. We then present the definition of the coverage of trajectory dataset over road network.

Definition 4 (Coverage of trajectory dataset over road network). Let E be the set of road segments, and let $D = \{T_1, T_2, \dots, T_{|D|}\}$ be a dataset of map-matched trajectories. Let R be the overall time period of interest, which is partitioned into non-overlapping unit time slots: $R = r_1 \cup r_2 \cup \dots \cup r_{|R|}$.

For each segment $e \in E$ and time slot $r \in R$, let $N_{r,e}(D)$ denote the number of trajectories in D that traverse segment e during time slot r (i.e., whose time interval intersect with r). Given a predefined coverage threshold θ , the utility of dataset D for road segment e

during time slot r is defined as $U_{r,e}(D) = \mathbb{I}(N_{r,e}(D) \geq \theta)$, where $\mathbb{I}(\cdot)$ is the indicator function that returns 1 if the condition holds, and 0 otherwise. The overall utility of dataset D is defined as the total number of sufficiently covered segment-time pairs: $U(D) = \sum_{e \in E} \sum_{r \in R} U_{r,e}(D)$.

2.2 Shapley-based valuation for trajectory dataset

To fairly assess the value of individual trajectory datasets, we adopt the Shapley value for data evaluation [4]. Shapley value is a classical concept from cooperative game theory which fairly quantifies each player's contribution. In our context, we define the Shapley value specifically for trajectory datasets, where the utility function is based on the coverage over road network.

Definition 5 (Shapley value for trajectory dataset). Given n trajectory datasets $D_1, D_2, \dots, D_n$, each dataset D_i consists of $|D_i|$ map-matched trajectories, and the utility function of the trajectory dataset $U(\cdot)$ are defined as its coverage over the road network, as stated in Definition 4. Then the Shapley value of dataset D_i is defined as:

$$SV_i = \sum_{S \subseteq N \setminus \{i\}} \frac{U(S \cup \{i\}) - U(S)}{n \cdot \binom{n-1}{|S|}}, \quad (1)$$

where N is the union of all datasets.

The Shapley value for trajectory dataset satisfies the following three fairness properties: null-player, symmetric-fairness, and linear-additivity, and is thus suitable for fair data evaluation.

2.3 Hardness of shapley-based valuation

In this part, we prove that the problem of trajectory Shapley value computation is #P-complete.

Lemma 1. Computing the Shapley value for n trajectory datasets is #P-complete.

Proof We prove the lemma by reducing the problem weighted voting games [5], a known #P-complete problem. Consider a special instance of Shapley value computation with a single road segment and a single time slot. Let each dataset D_i contribute a voting weight equal to its number of trajectories covering the segment, and set θ as the voting threshold in the weighted voting games. Then the utility function equals to 1 if the total coverage of the dataset coalition meets or exceeds θ , and 0 otherwise, which is equivalent to a weighted voting game. Thus, computing the Shapley value in this setting is #P-complete. $\square$

Given the high computational complexity of exact Shapley value calculation, our goal is estimate its value efficiently.

■ 3 Methodology

In this section, we present a clustering-based framework for efficient estimation, along with two optimization strategies to enhance the framework's performance.

3.1 Clustering-based framework for shapley-based valuation

To address the high computational cost of Shapley value estimation for trajectory datasets, we propose a clustering-based framework that

leverages the decomposable and redundant structure of the utility function. The key insight is that the overall Shapley value can be broken down into contributions from individual segment-time pairs, many of which exhibit similar behavior across datasets.

Specifically, we observe that the coverage over road network defined in Definition 4 is additive and can be decomposed across individual segment-time pair, we compute the coverage distribution across datasets as D_i can be expressed as the sum of, where play values over all segment-time pairs. This decomposition allows for more efficient computation by denotes the number of trajectories from dataset t , segment-time pairs that exhibit similar distributions of trajectory counts across datasets t that cover the segment-time pair.

- **Step 1: Coverage extraction.** For each trajectory dataset, compute its coverage on every segment-time pair, we compute the coverage distribution across datasets as $C = [c_1, c_2, \dots, c_n]$, where c_i denotes the number of trajectories from dataset D_i that cover the segment-time pair.
- **Step 2: Clustering.** Cluster all coverage vectors of segment-time pairs into k groups using furthest-point-first (FPF) algorithm. For each cluster, compute the Shapley value for its center.
- **Step 3: Value propagation.** Assign the Shapley value of the cluster center to all segment-time pairs within the same cluster. Then, we sum the contributions across all segment-time pairs to obtain the total Shapley value for each dataset.

3.2 Optimizations

In this part, we present two observation about the Shapley value for trajectory dataset. Based on these observations, we propose the optimization strategies within our clustering-based framework to reduce estimation error while keeping the number of segment-time pair Shapley value computations fixed.

Observation 1: Permutation invariance. The Shapley value is invariant under permutations of player indices, i.e., if we permute the indices of the datasets, their Shapley values are permuted in the same way. This property allows us to sort each coverage distribution vector C in ascending order before clustering. We apply the FPF algorithm to the sorted coverage vectors, compute the Shapley values for the cluster centers, and propagate them to the remaining members of the cluster. Sorting the vectors reduces intra-cluster variance, which improves the accuracy of the propagated Shapley values.

Observation 2: Saturation of coverage count. We further observe that once the coverage count for a dataset on a segment-time pair exceeds the threshold θ , any additional coverage does not affect the Shapley value computation. This is because such coverage always ensure the utility function $U(\cdot)$ evaluates to 1, regardless of the coalition of datasets. Therefore, we can safely clip any coverage value greater than θ to θ before clustering, without affecting the correctness of the Shapley value. This operation can reduces the variability in coverage distributions, enabling more compact clustering and thus improving the accuracy of value propagation.

4 Experiments

4.1 Experimental setup

We evaluate our method on two real-world trajectory datasets, **Beijing** [6], a dataset collected from five taxi companies (each regarded as a data source of its trajectories), and **Chengdu** [6], a dataset published by Didi Chuxing and partitioned into varying numbers of subsets to simulate the federated setting. We compare our method with three baselines: **Exact** [4], which compute the exact Shapley value according to Definition 5, **CCB** [7], a sampling-based estimation method for Shapley value with empirical Bernstein bound-based allocation strategy, and **IPSS** [8], a stratified sampling-based algorithm for Shapley value estimation. We evaluate all methods in terms of two metrics, running time and approximation error (the relative error in l_2 -norm).

All our code is implemented in Python, and all the experiments are conducted in a server with 24 2.60 GHz Intel(R) Xeon(R) Platinum 8361HC CPU processors and 128 GB of memory.

4.2 Experimental results

We conduct experiments on Beijing dataset by varying the time period length $|R|$ and the coverage threshold θ . As shown in Fig. 1, our method significantly consistently outperforms the baselines in all settings, achieving up to $15.2\times$ speedup in computation time. Besides, our method consistently achieves the lowest error among the approximate methods, demonstrating its accuracy in Shapley value estimation.

We perform the experiments on Chengdu dataset by varying the source number n . As shown in Fig. 2, our method achieve $4.8\times$ to $53.4\times$ speedup compared to existing solutions, while maintaining low

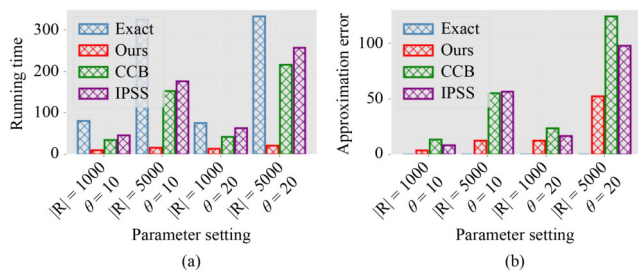


Fig. 1 Running time (a) and approximation error (b) of Shapley value computation for Beijing dataset with four parameter settings

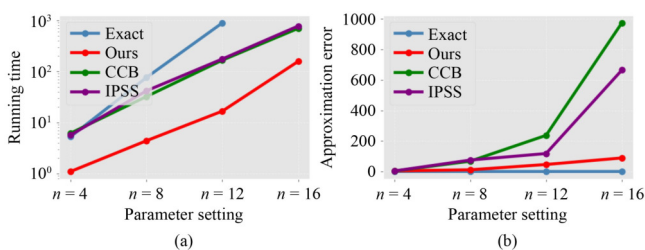


Fig. 2 Running time (a) and approximation error (b) of Shapley value computation for Chengdu dataset with different source number n

5 Conclusion

We define the problem of computing the Shapley value for trajectory datasets, enabling a fair evaluation of data value to incentivize sharing. To efficiently compute Shapley values, we propose a clustering-based framework that achieves up to a $15.2\times$ speedup and an $8.4\times$ reduction in approximation error compared to existing methods.

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Competing interests

Yongxin TONG is an Action Editor of the journal and a co-author of this article. To minimize bias, he was excluded from all editorial decision-making related to the acceptance of this article for publication. The remaining authors declare no conflict of interest.

References

- [1] Tong Y, She J, Ding B, Wang L, Chen L. Online mobile Micro-Task Allocation in spatial crowdsourcing. In: Proceedings of the 32nd IEEE International Conference on Data Engineering (ICDE). 2016, 49–60
- [2] Wei S, Tong Y, Zhou Z, Xu Y, Gao J, Wei T, He T, Lv, W. Federated reasoning LLMs: a survey. Frontiers of Computer Science, 2025, 19(12): 1912613.
- [3] Lou Y, Zhang C, Zheng Y, Xie X, Wang W, Huang Y. Map-matching for low-sampling-rate GPS trajectories. In: Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information. 2009, 352–361
- [4] Shapley L S. A value for N-person games. In: Kuhn H, Tucker A, eds. Contributions to the Theory of Games II. Princeton: Princeton University Press, 1953
- [5] Deng X, Papadimitriou C H. On the complexity of cooperative solution concepts. Mathematics of Operations Research, 1994, 19(2): 257–266
- [6] Wang Y, Zeng Y, Li S, Zhang Y, Zhou Z, Tong Y. Efficient and private federated trajectory matching. IEEE Transactions on Knowledge and Data Engineering, 2024, 36(12): 8079–8092
- [7] Zhang J, Sun Q, Liu J, Xiong L, Pei J, Ren K. Efficient sampling approaches to Shapley value approximation. Proceedings of the ACM on Management of Data, 2023, 1(1): 48
- [8] Wei S, Tong Y, Zhou Z, He T, Xu Y. Efficient data valuation approximation in federated learning: a sampling-based approach. In: Proceedings of the 41st IEEE International Conference on Data Engineering (ICDE). 2025, 2922–2934